

CLOUD COMPUTING COURSE

Lecture7-Compute Virtualization in CC

ABSTRACT

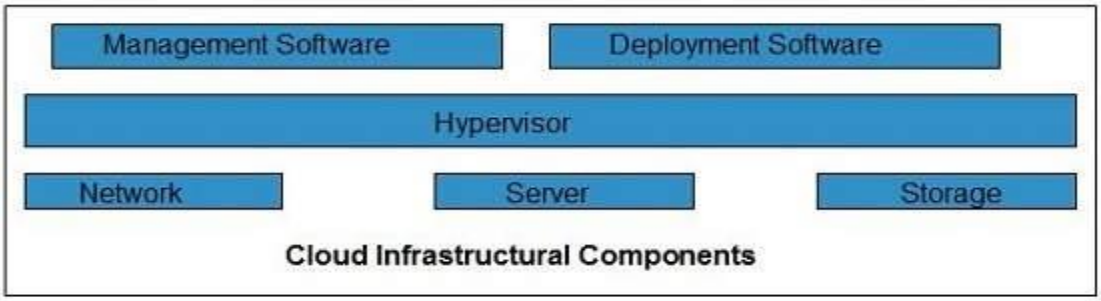
Lecture7 Introduces compute virtualization technology.

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Cloud Computing Infrastructure

Cloud computing infrastructure is the collection of hardware and software elements needed to enable cloud computing. It includes computing power, networking, and storage, as well as an interface for users to access their virtualized resources. The virtual resources mirror a physical infrastructure, with components like servers, network switches, memory and storage clusters.

Cloud infrastructure consists of servers, storage devices, network, cloud management software, deployment software, and platform virtualization.



1. **Management Software**

It helps to maintain and configure the infrastructure.

2. **Deployment Software**

It helps to deploy and integrate the application on the cloud.

3. **Hypervisor**

Hypervisor is a low-level program that acts as a Virtual Machine Manager. It allows to share the single physical instance of cloud resources between several tenants.

4. **Network**

It is the key component of cloud infrastructure. It allows to connect cloud services over the Internet. It is also possible to deliver network as a utility over the Internet, which means, the customer can customize the network route and protocol.

5. Server

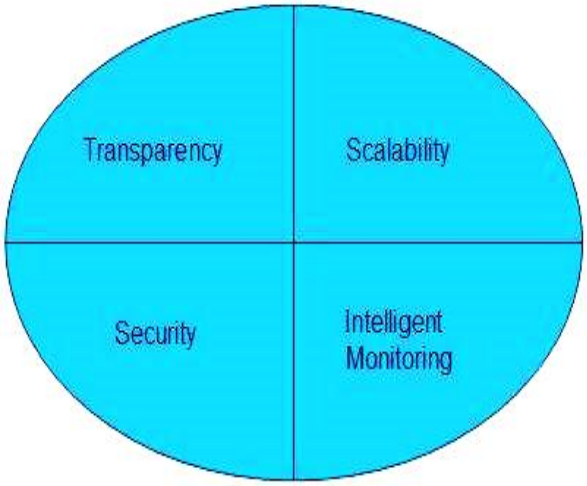
The server helps to compute the resource sharing and offers other services such as resource allocation and de-allocation, monitoring the resources, providing security etc.

6. Storage

Cloud keeps multiple replicas of storage. If one of the storage resources fails, then it can be extracted from another one, which makes cloud computing more reliable.

Infrastructural Constraints

Fundamental constraints that cloud infrastructure should implement are shown in the following diagram:



Transparency

It means that whether by the minute or second, you'll be able to look at the entire infrastructure in real time down to an individual VM, know how it is performing, and determine how this performance can impact your spending. You can track who has access to what resource, see how resources are being so that we can **scale them on demand**, issue **load balancing** and determine the **billing amount**.

Scalability

Scaling up an application delivery solution is not that easy as scaling up an application because it involves configuration overhead or even re-architecting the network. So, application delivery solution is need to be scalable which will require the virtual infrastructure such that resource can be provisioned and de-provisioned easily.

Intelligent Monitoring

To achieve transparency and scalability, application solution delivery will need to be capable of intelligent monitoring.

Security: The mega data center in the cloud should be securely architected.

Cloud Computing Technologies

The main enabling technology for cloud computing is **virtualization**.

Virtualization Definition:

Virtualization: is the creation of a virtual (rather than actual) version of something, such as an operating system (OS), a server, a storage device or network resources.

Virtualization History:

Virtualization began in the 1960s, as a method of logically dividing the system resources provided by mainframe computers between different applications.

Virtualization Benefits

Virtualization uses software that simulates hardware functionality in order to create a virtual system. This practice allows IT organizations to operate multiple operating systems, more than one virtual system and various applications on a single server. The benefits of virtualization include:

1. **Reduced IT costs:** reducing hardware servers, energy consumption, cooling power and reduce the ongoing administration and management of manual.
2. **Efficient resource utilization:** enables businesses to get the most out of their investment in hardware and resources.
3. **Easier testing**: Testing is less complicated in a virtual environment.
4. **Quicker backups:** Backups can be taken of both the virtual server and the virtual machine.
5. **Easier disaster recovery:** Disaster recovery is very simple in a virtualized environment, using snapshots and migration techniques.

Virtualization Types:

You probably know a little about virtualization if you have ever divided your hard drive into different partitions. A partition is the logical division of a hard disk drive to create, in effect, two separate hard drives.

There are **six areas** of IT where virtualization is making headway:

1. **Compute or Server virtualization** is the masking of server resources -- including the number and identity of individual physical servers, processors and operating systems -- from server users. The intention is to spare the user from having to understand and manage complicated details of server resources while increasing resource sharing and utilization and maintaining the capacity to expand later.
2. **Network virtualization** is a method of combining the available resources in a network by splitting up the available bandwidth into channels, each of which is independent from the others and can be assigned -- or reassigned -- to a particular server or device in real time. The idea is that virtualization disguises the true complexity of the network by separating it into manageable parts, much like your partitioned hard drive makes it easier to manage your files.
3. **Storage virtualization** is the pooling of physical storage from multiple network storage devices into what appears to be a single storage device that is managed from a central console. Storage virtualization is commonly used in storage area networks.
4. **Data virtualization** is abstracting the traditional technical details of data and data management, such as location, performance or format, in favor of broader access and more resiliency tied to business needs.
5. **Desktop virtualization** is virtualizing a workstation load rather than a server. This allows the user to access the desktop remotely, typically using

a thin client at the desk. Since the workstation is essentially running in a data center server, access to it can be both more secure and portable. The operating system license does still need to be accounted for as well as the infrastructure.

6. **Application virtualization** is abstracting the application layer away from the operating system. This way, the application can run in an encapsulated form without being depended upon on by the operating system underneath. This can allow a Windows application to run on Linux and vice versa, in addition to adding a level of isolation.

A key use of virtualization technology is **Compute (Server) Virtualization**

Let us start with compute virtualization technique…

